
Can Agent Benchmarks Support Their Scores?

Evidence-Supported Bounds for Interactive-Agent Evaluation

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Abstract

Interactive-agent benchmarks map an agent run to a binary outcome via outcome checks. When checks rely on surface-level signals or miss the agent’s action path, they fail to determine whether the run succeeded. For example, a benchmark case asks whether Alice’s shipping address changed, but its outcome check only verifies that the agent clicked Save. This does not guarantee that the intended state change was applied; the agent may have updated the wrong record. Counting such a run as a success makes the score misleading. Benchmark quality therefore depends on outcome detection as well as task design.

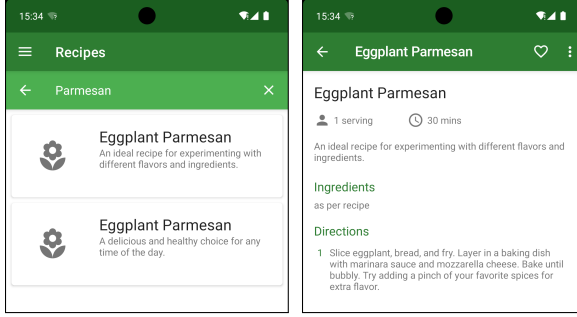
We address this by adding an outcome-evidence reporting layer to existing benchmarks, without changing their tasks, agents, or evaluators. The layer does three things: (i) specifies, before scoring, which stored artifacts would verify the claimed outcome for each case; (ii) applies the locked case checklist to each completed record and assigns one of three evidence labels: Evidence Pass, Evidence Fail, or Unknown; and (iii) reports evidence-supported bounds that quantify uncertainty from Unknown records. Unknown records are kept visible rather than silently counted, dropped, or hidden inside a single success rate.

We evaluate this outcome-evidence layer on four public benchmarks: AGENTDOJO, τ^3 -bench retail, APPWORLD, and WEBARENA-VERIFIED. The resulting reports make outcome-evidence quality visible alongside task quality, supporting more evidence-grounded agent evaluation and uncertainty reporting. We release the case checklists, reason labels, audit checks, robust-ranking summaries, and scoring script needed to reproduce the reports.

1 Introduction

Interactive-agent benchmarks increasingly ask agents to navigate interfaces, call tools, and change persistent environment state. Their headline result is usually a single success rate, but a reported success is only as reliable as the outcome check behind it. Figure 1 shows a simple example from the released ANDROIDWORLD benchmark: the task asks an agent to delete recipes whose directions contain Parmesan; the trace shows the agent opening visible Eggplant Parmesan recipes and deleting nothing; yet the benchmark still reports success. This kind of mismatch can arise when an evaluator is too weak, when a benchmark has a bug, or when the retained artifacts are insufficient to verify what happened.

This is an outcome-evidence gap. The benchmark claim is a binary statement about what happened in the environment, while the stored artifacts may include only screenshots, action logs, final messages, partial verifier inputs, or other indirect traces. The gap is especially important for interactive agents because correctness often depends on identity resolution, side effects, post-state queries, policy constraints, and multi-step tool interactions. When these artifacts are insufficient, a point score can overstate what the benchmark has actually verified. In an ideal benchmark whose claims, evaluators,



Screen 1: searched for Parmesan Screen 2: opened recipe details

Figure 1: One case from ANDROIDWORLD (source). The task asks the agent to delete recipes whose directions contain Parmesan. The retained screens show the key trace: the agent (i) searches for Parmesan, (ii) opens an Eggplant Parmesan recipe, and (iii) deletes nothing. The run is nevertheless reported as successful because the released target construction leaves the evaluator with an empty target set, while recipes treated as non-targets mention Parmesan. This creates a false success label: the benchmark output says the deletion succeeded.

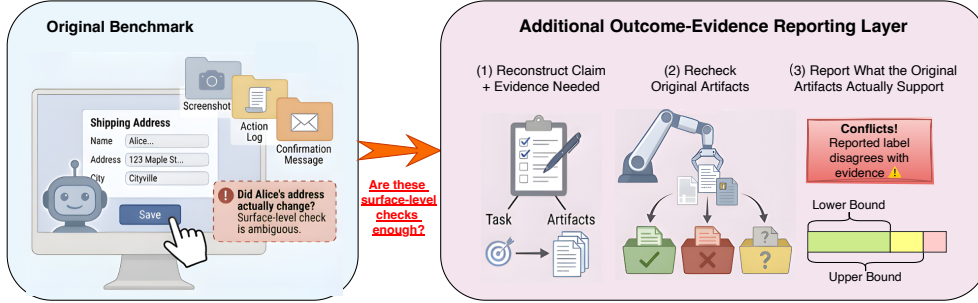


Figure 2: Overview of the outcome-evidence gap and our reporting layer. A benchmark may report success from surface artifacts even when those artifacts do not verify the intended state change. Our layer leaves the benchmark run unchanged, then states the claim and needed evidence, rechecks the benchmark’s saved artifacts, and reports what the original artifacts actually support: evidence labels, label conflicts, and evidence-supported bounds.

38 and retained state are fully aligned, every completed run would be decidable. Unresolved records
39 appear when the stored artifacts are insufficient to verify the benchmark’s own outcome claim.

40 We address this gap with an outcome-evidence reporting layer. The layer is additive: it leaves
41 benchmark tasks, agents, environments, and native evaluators unchanged. For each case, it writes
42 a short list of evidence needed before scoring, specifying which stored artifacts would decide the
43 benchmark’s own success claim. Each completed record is then labeled Evidence Pass, Evidence Fail,
44 or Unknown. Using P, F, U for the three counts, the report keeps Unknown records visible rather
45 than forcing them into success, failure, or exclusion.

46 When a record is Unknown, none of the usual shortcuts is neutral. Counting it as a success overstates
47 what the evidence supports. Counting it as a failure can punish missing instrumentation rather than
48 agent behavior. Dropping it changes which records are evaluated and can create agent-dependent post-
49 selection, because some agents leave more decisive traces than others. We therefore keep Unknown
50 records in the same set of completed records and report the range of all-record performance still
51 compatible with the stored evidence. The counted-only score is still useful, but only as a diagnostic
52 conditional on decidability, not as a headline all-record estimate.

53 We evaluate this reporting layer on four public interactive-agent benchmarks. AGENTDOJO and
54 τ^3 -bench retail provide evidence-sensitive settings where plausible traces can outrun stored evidence.
55 APPWORLD and WEBARENA-VERIFIED use strong state-based or verified evaluators and serve as
56 calibration controls. The study is designed to be falsifiable: large Unknown shares on the controls
57 would indicate over-conservative evidence requirements, while narrow control bounds and wider
58 evidence-sensitive bounds would show that the layer distinguishes weak stored evidence from strong
59 native verification.

The paper makes four contributions: (i) we formalize the outcome-evidence gap in interactive-agent benchmarks, where a binary success claim may be reported even when stored artifacts are insufficient to decide that claim; (ii) we introduce case checklists tied to each benchmark’s own success claim, plus a three-state counting rule that separates Evidence Pass, Evidence Fail, and Unknown records without changing benchmark tasks, agents, environments, or native evaluators; (iii) we derive evidence-supported performance bounds over the fixed set of completed records, making the counted-only score a diagnostic over decidable records rather than a headline all-record estimate; and (iv) we apply the method to four public benchmarks, showing that evidence uncertainty is concentrated in evidence-sensitive domains while remaining narrow on strong state-based evaluator controls, and release the case checklists, reason labels, audit checks, robust-ranking summaries, and scoring script needed to reproduce the reports.

2 Related Work and Preliminaries

Interactive-agent evaluation and outcome checks. Interactive-agent benchmarks evaluate agents that navigate interfaces, call tools, and alter environment state. Web-agent benchmarks include WebShop [27], Mind2Web [3], VisualWebArena [9], WebArena [30], and BrowserGym [1]. Tool-use and stateful-interaction benchmarks include ToolBench-style tool-use benchmarks [18], τ -bench [28], τ^3 -bench retail [21], and ToolSandbox [10]. Broader agent environments include AGENTDOJO [2], ANDROIDWORLD [19], WORKARENA [4], OSWORLD-VERIFIED [24], and long-horizon workplace settings such as TheAgentCompany [25]. These benchmarks make interactive evaluation realistic, but realism alone does not guarantee that a stored trace decides the outcome claim being reported.

Verified evaluators and benchmark validity. A recent line of work strengthens outcome checks through state-based or verified evaluation, including the verified WEBARENA-VERIFIED release [7], APPWORLD database-state unit tests [22], and SWE-bench Verified [15]. Benchmark-validity work, including the Agentic Benchmark Checklist, argues that construct validity, outcome validity, and reporting assumptions should be made explicit [31]. Our contribution is complementary: rather than proposing a new environment, a stronger domain evaluator, or an audit checklist, we add a cross-benchmark reporting layer that asks what the stored artifacts of completed runs can support under each benchmark’s own claim. LLM-as-judge and trajectory-evaluation work [29, 20, 11], benchmark refresh or contamination studies [8, 6, 26], and empirical audits of benchmark solutions [23, 16] are useful diagnostics, but they do not replace evidence that a native outcome claim is decidable from stored artifacts. Dataset and model reporting work similarly argues that evaluation artifacts should make their assumptions visible [5, 14, 12, 17]; our unit is a completed interactive-agent record and the artifacts retained to verify its outcome claim.

Claims, records, traces, and evidence. A *case* specifies a task, the initial environment, the benchmark claim being tested, and the evidence required to decide that claim. A benchmark claim is the binary outcome statement the benchmark reports for a completed run, such as whether a target record was updated, a message was sent, or a constraint was violated. The *native evaluator* is the benchmark’s released mechanism for turning its available artifacts into a success or failure label.

We use *case unit* only for a small bookkeeping issue: in most domains, one task instance is one case unit, while in paired-arm settings such as AGENTDOJO, two runnable arms form one case unit even though they are executed separately. Decisions live at the level of *records*. A record is one agent’s result on one case unit. The distinction matters because the same case unit can be countable for one agent, whose trace exposes decisive evidence, and Unknown for another, whose trace does not.

An *episode* is one concrete execution of one agent on one runnable arm of a case. Episodes produce *traces*: ordered records of actions, observations, tool calls, browser events, messages, files, and post-run artifacts. We use *stored artifacts* for the full set of traces, logs, state snapshots, receipts, verifier inputs, and post-run measurements retained after execution. *Evidence* is the subset of those artifacts that can support the active benchmark claim. A trace can be plausible without being decisive.

Staying tied to the benchmark claim. Because the method sits on top of existing benchmarks, the requirements must not silently make the task harder. For each case unit, we write down: (i) the benchmark’s own binary claim; (ii) the official source of that claim, such as the task text, evaluator

code, policy, or schema; (iii) the artifacts needed to decide success or failure; (iv) the rule for assigning Unknown; and (v) whether any requirement goes beyond the benchmark’s own claim.

The requirements must be minimal with respect to the benchmark’s own semantics. For example, if a task says only “update the address” and the native evaluator only checks the target record’s address, then the requirements ask for target-record identity and post-state evidence. A no-collateral-change requirement is included only if the benchmark’s own task, policy, evaluator, or reported claim requires it. Otherwise it is labeled as a stronger measurement claim and reported separately.

When sources differ, the requirements follow a fixed source hierarchy: official evaluator semantics first; official task text and policy second; and schema constraints needed to interpret evaluator-visible state third. We do not add requirements from annotator intuition alone. Any requirement not grounded in one of these sources is labeled as a stronger measurement claim and excluded from the main bounds tied to the benchmark’s own claim.

3 Method

The method turns completed benchmark runs into an evidence-supported report. It keeps the benchmark’s native view, namely the released evaluator’s pass/fail label, and adds an evidence view that asks whether the stored artifacts are sufficient to decide the same underlying outcome claim. The layer is post-run with respect to agent execution, but fixed before evidence scoring: it does not change the task, prompt, agent, environment, native evaluator, or native score.

What Unknown means. Unknown is not a third kind of agent behavior. A record receives the Unknown label when the available artifacts do not decide whether the benchmark’s own success claim is true. A clicked Save button, a final confirmation message, or a passed proxy check may be insufficient if the retained state does not show that the intended record changed. Conversely, timeouts, invalid actions, tool misuse, and aborts after task start are failures when the native benchmark would fail them; they are not moved into Unknown. In an ideal state-based benchmark with explicit claims, aligned evaluators, complete retained state, and unique entity mappings, Unknown records should be rare or absent. Their presence measures an evidence gap in the benchmark report, not task difficulty or agent uncertainty.

Case checklist. For each case unit, we first make the outcome claim explicit. An LLM-assisted drafter reads the user goal, task text, official policy when present, released evaluator or oracle code, and relevant state schemas. It proposes a compact case checklist: what the user is trying to achieve, what condition the benchmark reports as success, which official code or oracle checks that condition, which artifacts would decide it after execution, and when the record should remain undecided. For every checklist field filled by the LLM-assisted drafter, the checklist must include either a rationale or a source pointer identifying the file and location from which the information was derived. Human reviewers source-check each checklist against the hierarchy in Section 2. The checklist is not a new task or answer key; it states what stored evidence would be sufficient to verify the benchmark’s own claim. In addition, the LLM-assisted drafter may propose conditions that would require a stronger measurement claim, together with a brief rationale for why those conditions go beyond the benchmark’s native success criterion. Human reviewers then adjudicate whether each proposed stronger measurement claim is reasonable and sufficiently supported. We report results in two layers: the native-aligned result, which follows the benchmark’s original claim, and a separate stronger-measurement result, which applies the additional reviewed conditions.

Evidence scoring. We then run the benchmark normally and retain the artifacts available for review: traces, tool calls, evaluator inputs and outputs, final messages, and saved database, browser, file, or API state. For each completed record, the native view records the released evaluator’s label. The evidence view applies the locked case checklist to the stored artifacts and assigns one of three labels: Evidence Pass, if the artifacts support success under the benchmark claim; Evidence Fail, if they support failure; and Unknown, if they do not decide the claim. For every judgment made or proposed by an LLM-assisted scorer, the record must include either a rationale or a source pointer identifying the file and location from which the information was derived. When a checklist includes reviewed stronger-measurement conditions, we report the evidence results in two layers: a native-aligned result, which follows the benchmark’s original claim, and a separate stronger-measurement result, which

163 applies the additional reviewed conditions. Thus the evidence view is an additional report over the
 164 same runs, not a replacement for the native benchmark score.

165 **Counting rule.** The reporting rule uses only the three evidence-state counts. For agent a in domain
 166 d , let $N_{a,d}$ be the number of completed records in the evaluation set, and let $P_{a,d}$, $F_{a,d}$, and $U_{a,d}$ be
 167 the counts of Evidence Pass, Evidence Fail, and Unknown records, so $N_{a,d} = P_{a,d} + F_{a,d} + U_{a,d}$.
 168 The counted-only score answers a conditional question: among records the stored artifacts decide,
 169 what fraction are successes? The bound answers the all-record question: over every completed record,
 170 what performance values are still compatible with the stored evidence? Writing P, F, U, N for the
 171 counts in a given agent-domain cell, we report

$$\text{CountedScore}_r(a, d) = \frac{P}{P + F}, \quad \text{Perf}_r(a, d) \in \left[\frac{P}{N}, \frac{P + U}{N} \right], \quad \text{Width}_r(a, d) = \frac{U}{N}.$$

172 The lower endpoint counts only Evidence Pass records; the upper endpoint treats every Unknown
 173 record as a possible success. The width is exactly the Unknown share. When $U = 0$, the bound
 174 collapses to the all-record score. When $U > 0$, a single point estimate must make an unobserved
 175 decision about records the artifacts did not decide. These intervals are not confidence intervals; they
 176 are partial-identification bounds describing what the stored evidence can and cannot identify [13].
 177 The counted-only score remains useful as a diagnostic conditional on decidability, but the headline
 178 object is the evidence-supported bound.

179 **Comparisons and diagnostics.** We keep three diagnostics separate from the headline bounds. First,
 180 case-cluster bootstrap intervals describe sensitivity to the finite set of sampled cases. Each replicate
 181 resamples case units within a domain and keeps all agent records attached to the resampled case
 182 unit. These intervals condition on the stored traces and fixed agent versions; they do not measure
 183 uncertainty from repeated executions, API drift, tool nondeterminism, or environment volatility.

184 Second, pairwise rankings are supported only when the bounds separate. For agents i and j in
 185 domain d , the bound-implied difference interval is $\Delta(i, j; d) = [\text{LB}_{i,d} - \text{UB}_{j,d}, \text{UB}_{i,d} - \text{LB}_{j,d}]$. We
 186 report $i \succ j$ only if $\text{LB}_{i,d} > \text{UB}_{j,d}$, and $j \succ i$ only if $\text{LB}_{j,d} > \text{UB}_{i,d}$. Otherwise the pair is not
 187 identified and is marked “?” rather than a tie. For a separated ordering $i \succ j$, the dominance margin
 188 is $M_{i \succ j} = \text{LB}_{i,d} - \text{UB}_{j,d}$. We add an asterisk when the 2.5th percentile of the case-cluster bootstrap
 189 distribution of $M_{i \succ j}$ remains positive; it marks descriptive stability, not significance.

190 Third, every U record receives one reason, so the bound width can be audited. The reasons are:
 191 missing state query (needed backend or environment state was not saved), unobservable side effect
 192 (the consequence of an action is not exposed), ambiguous identity mapping (the acted-on record
 193 or recipient is not unique), missing state-preservation evidence (the native claim requires unrelated
 194 state to remain unchanged but no diff or snapshot is available), paired-arm asymmetry (one arm of
 195 a paired case is decidable and the other is not), evaluator-output ambiguity (the released evaluator
 196 emits a label but the stored inputs do not determine it), and claim-scope mismatch (the trace supports
 197 a related task but not the reported claim). Appendix C gives representative examples and the priority
 198 rule used when more than one reason could apply.

199 4 Experiments

200 We evaluate whether the reporting layer changes what benchmark scores can claim, not which model
 201 wins a leaderboard. The experiment fixes four public interactive-agent benchmarks, three current
 202 frontier agents, and a 100-case sample per benchmark. Each run is scored twice: once by the
 203 benchmark’s native evaluator and once by the evidence-supported reporting layer.

204 4.1 Benchmarks

205 The benchmark suite contains two strong-evaluator controls and two domains where the native claim
 206 can depend on evidence that is not always stored. This mix is important: a useful reporting layer
 207 should leave strong native verification mostly narrow while exposing uncertainty when traces are
 208 insufficient. Across domains, the evidence view uses the artifacts retained by the normal benchmark
 209 run, including traces, tool calls, evaluator inputs and outputs, final messages, and saved database,
 210 browser, file, or API state when available.

Table 1: Benchmark suite. Each benchmark contributes 100 case units and is run with three agents. The suite pairs two evidence-sensitive domains with two strong-evaluator controls.

Benchmark	Study role	Native check
τ^3 -bench retail	Evidence-sensitive	Retail-state success signal.
AGENTDOJO	Evidence-sensitive	Paired utility/security labels.
APPWORLD	Control	Database-state unit tests.
WEBARENA-VERIFIED	Control	Verified deterministic evaluator.

4.2 Agents and workload

We run three agents through OpenRouter: Agent A is OpenAI GPT-5.4, Agent B is Claude Opus 4.7, and Agent C is DeepSeek V4 Pro. All three use temperature 0, a 4096-token output budget, a 120-second timeout, and two retries. The paper does not treat these models as an exhaustive comparison set. They are fixed measurement probes used to test whether outcome-evidence uncertainty is a property of benchmark evidence rather than an artifact of one agent family. The release records concrete model identifiers, prompts, tool wrappers, run dates, and API settings.

For each benchmark, the manifest selects 100 eligible case units using a deterministic hash order after excluding smoke-test and malformed cases. Each case unit is run once per agent, giving $4 \times 100 \times 3 = 1200$ completed record slots before infrastructure exclusions. AGENTDOJO executes two arms per record, so the main experiment stores 1500 execution episodes in total: 600 for AGENTDOJO and 300 for each of the other three benchmarks.

4.3 Execution and evidence scoring

Every agent run uses the benchmark’s original task interface, tools, environment, and released evaluator. Evidence collection is handled by benchmark-specific adapters that save raw run records, artifact manifests, native evaluator outputs, traces, logs, and available post-run state. The adapters do not make final evidence-label decisions, so the native score remains directly recoverable from the saved artifacts.

Before evidence scoring, an LLM-assisted drafter produces one case checklist per case unit. Drafting uses OpenAI GPT-5.4 through OpenRouter with temperature 0, high reasoning effort, a 12,000-token output budget, and a 180-second timeout. The drafter receives the user goal, task text, official policy when present, evaluator code or description, relevant database/API/browser/file/tool schemas, and the schema of artifacts saved after the run. Each LLM-filled checklist field must carry either a rationale or a source pointer to the file and location that supports it. Two researchers cross-validate the checklists in two review rounds. They check whether each checklist follows the source hierarchy in Section 2; unsupported items are manually edited or regenerated and reviewed again. Once locked, the checklists cannot be changed based on agent outcomes. If the drafter proposes stronger-measurement conditions, reviewers adjudicate them separately and score them outside the native-aligned result.

Final evidence scoring applies the locked checklist to each saved evidence directory using Codex CLI with GPT-5.4, xhigh reasoning effort, a read-only sandbox, and at most two attempts. The scoring bundle stores the score file, manifest, stdout/stderr logs, telemetry, event stream, and reasoning summary so that each Evidence Pass, Evidence Fail, or Unknown judgment can be audited.

Denominator rule. Only infrastructure or pre-run failures are excluded from the evaluation set. Examples include a sandbox that fails to boot before the task is presented, a missing benchmark asset, or an external service outage that prevents normal benchmark execution. Agent-caused invalid actions, tool misuse, timeouts after task start, malformed final answers, or benchmark-facing aborts are not excluded. They are counted as failures when the native benchmark would fail the run, and they are also reported in the all-attempt audit in Table 6. This separation prevents evidence filtering from raising scores by removing difficult or agent-induced failures.

Sample size and precision. The primary unit for uncertainty is the case unit, not the record. Each benchmark has 100 case units crossed with three agents, and the three records attached to a case unit can be correlated because the agents face the same task. We therefore avoid formal precision

Table 2: Main measurement report. Counts are over completed records after excluding only infrastructure/pre-run failures. Native score is the released benchmark score on the same completed records. Coverage is the share of records that stored artifacts decide. Counted-only score is conditional on those decidable records. Lower and Upper are the all-record evidence-supported endpoints; Width is the Unknown share.

Benchmark	Total	Native score	E-Pass	E-Fail	Unknown	Coverage	Counted-only score	Lower	Upper	Width
τ^3 -bench retail	300	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
APPWORLD	300	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
WEBARENA-VERIFIED	300	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
AGENTDOJO	300	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
All benchmarks	1200	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL

claims based on 300 independent records per benchmark. A record-level binomial calculation would be, at best, an independence lower bound and is not used for inference. All resampling intervals are case-clustered and condition on the stored traces.

4.4 Audit checks

We run two checks for inspectability. First, two researchers audit 100 records, 25 per benchmark. The sample is stratified within each benchmark to include decidable records, Unknown records, and native/evidence disagreements when all three buckets are present. Auditors see the task, trace, artifacts, and locked checklist, but not agent identity, native label, evidence label, or Unknown reason.

Second, we rerun a fixed reference agent, OpenAI GPT-5.4, on ten case units per benchmark. This rerun subset checks whether evidence labels are stable under a small repeated-execution probe. It is not used as an uncertainty interval for agent performance.

Adoption note. Adopting the method on top of an existing benchmark requires writing one fixed claim and one case checklist per case unit, then running the scoring script over stored traces. The script returns Evidence Pass, Evidence Fail, and Unknown counts, coverage, counted-only score, lower and upper bounds, bound width, and Unknown-reason records.

5 Evaluation

The evaluation is organized around the questions a reader of an agent benchmark actually needs answered. What did the original benchmark report? How much of that report is directly supported by stored artifacts? If support is missing, how large is the uncertainty, where does it come from, and does it change model comparisons? The tables below specify exactly what evidence a final benchmark report should expose. We interpret the Unknown share as a property of stored benchmark evidence, not as an agent error rate.

5.1 Evaluation 1: Evidence support for reported scores

The first result is the score report itself. Table 2 keeps the native benchmark score visible, then adds the evidence view: Evidence Pass, Evidence Fail, and Unknown counts, coverage, the counted-only score, and the lower and upper endpoints. This table should let a reader tell the difference between two very different situations: a native score that is backed by nearly complete stored evidence, and a native score whose apparent precision comes from treating Unknown records as if they had been decided. Unless otherwise marked, the main tables report the native-aligned layer. Stronger-measurement results are reported separately in Appendix D so that the paper does not silently turn a benchmark into a stricter benchmark.

5.2 Evaluation 2: Counted-only scores are conditional on decidability

The counted-only score answers a narrow but useful question: among records where stored artifacts decide the benchmark claim, how often is success supported? Its denominator is therefore not the full

Placeholder: native score and evidence-supported endpoints.

Final figure: one row per benchmark and model. Plot the native score, the counted-only diagnostic score, and the evidence-supported interval [Lower, Upper]. Sort rows by Width.

Figure 3: Planned visual summary of Table 2 at the model-by-benchmark level. The figure should make the paper’s main empirical pattern visually falsifiable: strong state-based benchmarks should have narrow intervals, while evidence-sensitive benchmarks may have wide intervals.

Table 3: Main Unknown-evidence diagnostic. Width is the Unknown share from Table 2. The top reason identifies the largest source of missing evidence; the final column should be filled with the concrete benchmark repair suggested by that reason.

Benchmark	Width	Top Unknown reason	Benchmark repair suggested by the evidence gap
τ^3 -bench retail	FILL	FILL	FILL
APPWORLD	FILL	FILL	FILL
WEBARENA-VERIFIED	FILL	FILL	FILL
AGENTDOJO	FILL	FILL	FILL

evaluation set when Unknown records exist. In the final analysis, a high counted-only score with low coverage should be described as “high success on the decidable subset,” not as strong all-record performance. Case-resampling intervals for the same quantities appear in Appendix D.

The next diagnostic asks why records are Unknown. Table 3 should not be read as a new score; it is a repair map for benchmark authors. If most Unknown records come from missing state queries, the fix is better post-run state capture. If they come from identity ambiguity, the fix is unique identifiers or stricter task schemas. If they come from claim-scope mismatch, the benchmark needs a clearer statement of what its score claims.

5.3 Evaluation 3: Model rankings require separated bounds

The third result asks whether leaderboard comparisons survive Unknown records. Table 4 reports a directional ordering only when one model’s lower endpoint exceeds another model’s upper endpoint. A question mark means the ordering is not identified from stored evidence; it is not a tie. This is deliberately stricter than comparing native point scores, because the point scores may differ only inside the Unknown part of the evaluation set.

5.4 Evaluation 4: Counting decisions are inspectable

The fourth result checks that the labels are not just another opaque judge. The blinded audit samples counted records, Unknown records, and native/evidence disagreements. The rerun subset checks whether labels remain stable on a small repeated-execution probe. Disagreements are useful: they identify cases where the requirement text, artifact capture, or reason taxonomy needs clarification.

6 Discussion

The central empirical interpretation should be filled from Table 2 and Figure 3. In the final version, this paragraph should state which benchmark has the largest evidence gap, whether the strong-evaluator controls have narrow bounds, and whether native scores fall inside the evidence-supported endpoints: FILL. This is the main falsification check for the paper. If strong state-based controls have wide bounds, the case checklists are too conservative or the stored artifacts are weaker than expected; if evidence-sensitive domains have wide bounds while controls remain narrow, the method is detecting a real outcome-evidence difference rather than mechanically lowering scores.

The method is deliberately narrower than a new evaluator. It does not change tasks, prompts, environments, agents, or the native score. Its claim is that a benchmark report should distinguish what the released evaluator counted from what stored artifacts can support. The discussion of rankings

Table 4: Directional pairwise robust-ranking matrix. Three agents induce three pairwise comparisons per benchmark. A cell $A > B$ is reported only when $LB_{A,d} > UB_{B,d}$; ? denotes bound overlap. An asterisk marks unadjusted case-resampling stability, not statistical significance. It is added only when the observed bounds are separated and the lower 2.5th percentile of the case-cluster bootstrap dominance margin remains positive.

Benchmark	GPT-5.4–Claude 4.7	GPT-5.4–DeepSeek V4 Pro	Claude 4.7–DeepSeek V4 Pro	Non-identified pairs
τ^3 -bench retail	?	?	?	FILL
APPWORLD	?	?	?	FILL
WEBARENA-VERIFIED	?	?	?	FILL
AGENTDOJO	?	?	?	FILL
All benchmarks	Unsupported native strict orderings: FILL; native conflicts: FILL			FILL

Table 5: Stratified audit and rerun checks. Audit strata report counted records, Unknown records, and native/evidence disagreements. Agreement and κ values are descriptive for this stress sample, not unweighted population estimates. Taxonomy κ uses {R1–R7} on items both auditors marked Unknown. Rerun agreement checks label stability only.

Benchmark	Audit items	Audit strata	Audit agreement	Countability κ	Taxonomy κ	Rerun agreement / dominant disagreement
τ^3 -bench retail	25	FILL	FILL	FILL	FILL	FILL; FILL
APPWORLD	25	FILL	FILL	FILL	FILL	FILL; FILL
WEBARENA-VERIFIED	25	FILL	FILL	FILL	FILL	FILL; FILL
AGENTDOJO	25	FILL	FILL	FILL	FILL	FILL; FILL
All benchmarks	100	FILL	FILL	FILL	FILL	FILL; FILL

should therefore be filled from Table 4: FILL. A non-identified ordering is not a tie; it means the stored evidence supports more than one completion of the Unknown records.

The bounds are also not confidence intervals. They are induced by Unknown evidence. Case-cluster bootstrap intervals answer a separate question about finite-case resampling conditional on stored traces and fixed agent versions. The audit and rerun checks should be summarized here once filled: FILL. Their role is inspectability, not a repeated-execution performance estimate.

The main limitations follow from the design. A vague benchmark claim cannot be rescued by a cleaner report; the seven-category Unknown-reason taxonomy is a study vocabulary rather than a universal ontology; three agents are measurement probes rather than a definitive leaderboard; and stronger-measurement conditions must remain separate from the native claim. The practical implication is direct: benchmark reports should include Evidence Pass, Evidence Fail, and Unknown counts, coverage, counted-only score, lower and upper endpoints, width, and reason labels. A benchmark can make the bounds collapse by retaining the state needed to verify its own claim. Unknown records are therefore not an unavoidable feature of interactive-agent evaluation; they are a diagnostic signal about what a benchmark report does and does not substantiate.

7 Conclusion

Interactive-agent benchmarks often report scores over executed tasks, but execution is not the same as evidence-supported measurement. We introduced a reporting layer tied to each benchmark’s own claim: it classifies completed records as Evidence Pass, Evidence Fail, or Unknown, then reports the evidence-supported bounds implied by Unknown records. Across AGENTDOJO, APPWORLD, WEBARENA-VERIFIED, and τ^3 -bench retail, the evaluation asks whether uncertainty is visible where stored artifacts do not decide the benchmark claim and remains narrow where native evaluators already provide strong state-based evidence. Future benchmark reports should include Evidence Pass, Evidence Fail, Unknown, and bounds, so that a score is read together with the evidence that can actually support it.

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456 A Denominator Audit

Table 6: Denominator audit. Only infrastructure or pre-run failures are excluded before forming the completed evaluation set. Agent-caused aborts, invalid actions, and timeouts after task start stay in the denominator as native failures.

Domain	Attempted	Infra/pre-run excluded	Completed	Agent-caused fail	Notes
AGENTDOJO	FILL	FILL	300	FILL	Paired benign/injected arms remain linked at the case-unit level.
APPWORLD	FILL	FILL	300	FILL	Native database-state failures remain in the evaluation set.
WEBARENA- VERIFIED	FILL	FILL	300	FILL	Browser-task timeouts after start remain in the evaluation set.
τ^3 -bench retail	FILL	FILL	300	FILL	Tool misuse and policy-path failures remain in the evaluation set.
All domains	FILL	FILL	1200	FILL	Exclusions are not allowed to depend on whether the agent likely succeeded.

457 B Case Checklists and Scoring Inputs

458 Before scoring, we create one fixed case checklist per sampled case unit. The checklist is drafted
 459 with an LLM and then reviewed by two researchers in two cross-validation rounds. The draft records
 460 the user’s goal, what the official benchmark oracle or evaluator checks, what actions or state changes
 461 are needed for completion, and which stored artifacts would let a scorer decide the benchmark’s
 462 own claim. Every LLM-filled field must include either a rationale or a source pointer to the file
 463 and location from which the information was derived. If a draft item is unsupported, ambiguous, or
 464 stronger than the native claim, reviewers edit it or regenerate that item and review it again. Proposed
 465 stronger-measurement conditions are adjudicated separately and never folded into the native-aligned
 466 result without being marked. The checklist is locked before agent identities, agent traces, native
 467 scores, or evidence labels are shown to the scorer.

468 After each run, we collect the stored artifacts and score the record twice: the native benchmark
 469 evaluator gives the official/native label, and our scorer uses the locked checklist to assign Evidence
 470 Pass, Evidence Fail, or Unknown. Any LLM-assisted scoring judgment must also carry a rationale
 471 or source pointer. This separation is important: the paper reports both what the original benchmark
 472 would have counted and what the stored artifacts can support under the additional case checklist.

Table 7: Fields in the locked case checklist. The checklist is a scoring aid, not a new task.

Field	Purpose	Required support
User goal	Restates the task in benchmark language, including the intended object or recipient when available.	Task file, policy text, or rationale when the goal is synthesized across sources.
Native success claim	States what the released benchmark score claims to measure for this case.	Evaluator/oracle pointer, task text pointer, or policy pointer.
Official oracle or evaluator	Points to the official unit test, verifier, reward function, policy rule, or evaluator output used by the benchmark.	File path and code location, or released evaluator documentation.
Completion requirements	Lists the minimal state changes or side effects needed for the native claim to be true.	Source pointer for each requirement; rationale only when derived from multiple cited sources.
Evidence needed	Lists stored artifacts that can verify the claim, such as state diffs, database rows, receipts, verifier inputs, or unique identifiers.	Artifact schema pointer or rationale connecting the artifact to the native claim.
Evidence-label rule	Gives the scorer a deterministic rule for assigning Evidence Pass, Evidence Fail, or Unknown from stored artifacts.	Source pointer or rationale for each branch of the rule.
Stronger-measurement conditions	Separates reviewed non-native checks from the native-aligned result.	Brief rationale explaining why the condition goes beyond the native success criterion, plus reviewer adjudication.
Review notes	Records reviewer edits, regeneration reason, and final approval in two cross-validation rounds.	Reviewer id, round id, and decision note.

Table 8: Case-checklist drafting metadata. These values are filled from the scored manifest and released with the review package.

Field	Value
Drafter provider and model	OpenRouter; OpenAI GPT-5.4
Temperature / reasoning effort	0; high
Output budget / timeout	12,000 tokens; 180 seconds
Prompt version	FILL
Visible inputs	Task text; official policy, if any; evaluator code or evaluator description; database/API/browser/file/tool schema; trace schema; available post-run artifact types
Hidden inputs	Agent identity; agent trace; native score; outcome label; alternate-view verdicts
Human review rule	Two researchers cross-validate every checklist in two rounds before scoring. Unsupported requirements are edited or regenerated, then rechecked. Requirements that go beyond the benchmark claim are adjudicated separately and kept out of the native-aligned result.
Source hierarchy	Evaluator semantics first; task text and policy second; schema constraints needed to interpret evaluator-visible state third.

473 C Unknown-Reason Taxonomy

474 Each Unknown record receives one upstream blocking reason. The labels make missing evidence
475 actionable for benchmark authors; they are not a universal ontology.

Table 9: Unknown-reason labels. The upstream-priority rule assigns the reason that, if resolved first, would unblock the other missing evidence.

Reason	What is missing	Representative example
R1: Missing state query	The benchmark claim depends on a back-end or environment state value that was never queried or stored.	A database task has no post-run read of the relevant table.
R2: Unobservable side effect	The action is performed, but delivery, send, payment, receipt, or another side effect is not exposed in any artifact.	A message appears to be sent, but the tool returns no receipt or message id.
R3: Ambiguous identity mapping	Multiple records or recipients match the claim and the trace does not identify which one was acted on.	Two customers share the same name and no unique id is logged.
R4: Missing state-preservation evidence	The native task or policy requires preserving unrelated state, but no snapshot or diff can check it.	A calendar edit requires preserving other events, but no post-run diff is captured.
R5: Paired-arm asymmetry	A paired case requires both arms to be decidable, but one arm is decidable and the other is not.	The benign arm is verifiable while the injected arm’s security claim is not.
R6: Evaluator-output ambiguity	The released evaluator emits a label, but its stored inputs do not uniquely support that label under the benchmark claim.	A verifier accepts a string match without evidence that the structured target state was reached.
R7: Claim-scope mismatch	The trace supports a related task, but not the benchmark claim being reported.	The agent gives a helpful refund response, but the policy-relevant backend update required by the claim is absent.

476 D Additional Evaluation Tables

477 The main text keeps only the tables needed to follow the argument. This section holds the longer
478 tables that will be filled directly from the scored manifest.

Table 10: Per-agent evidence-supported bounds by benchmark. Each cell reports native score, counted-only score, lower endpoint, upper endpoint, and width.

Benchmark	OpenAI GPT-5.4	Claude 4.7	DeepSeek V4 Pro
AGENTDOJO	FILL	FILL	FILL
APPWORLD	FILL	FILL	FILL
WEBARENA-VERIFIED	FILL	FILL	FILL
τ^3 -bench retail	FILL	FILL	FILL

Table 11: Stronger-measurement results. These rows apply only reviewed conditions beyond the benchmark’s native success criterion and are reported separately from Table 2.

Benchmark	Stronger condi- tion family	E-Pass	E-Fail	Unknown	Coverage	Lower	Upper	Width	Difference from native-aligned result
AGENTDOJO	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
APPWORLD	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
WEBARENA-VERIFIED	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
τ^3 -bench retail	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL

Table 12: Descriptive case-resampling intervals for evidence-supported quantities. These intervals condition on stored traces and fixed agent versions; they do not include repeated-execution randomness, API drift, tool nondeterminism, or environment volatility.

Domain	Lower endpoint	Upper endpoint	Width	Counted-only score	Native score inside bounds
AGENTDOJO	FILL	FILL	FILL	FILL	FILL
APPWORLD	FILL	FILL	FILL	FILL	FILL
WEBARENA-VERIFIED	FILL	FILL	FILL	FILL	FILL
τ^3 -bench retail	FILL	FILL	FILL	FILL	FILL
All domains	FILL	FILL	FILL	FILL	FILL

Table 13: Full Unknown-reason distribution. The main text reports the top reason per benchmark; this table exposes the complete reason histogram.

Benchmark	R1	R2	R3	R4	R5	R6	R7	Interpretation
AGENTDOJO	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
APPWORLD	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
WEBARENA-VERIFIED	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL
τ^3 -bench retail	FILL	FILL	FILL	FILL	FILL	FILL	FILL	FILL

Table 14: Pairwise dominance margins for bound-separated comparisons. For a reported ordering $i > j$, the margin is $M_{i>j} = \text{LB}_{i,d} - \text{UB}_{j,d}$. The bootstrap interval is a descriptive stability check, not a formal significance test.

Domain	Pair	Bound decision	Margin and case-resampling interval	Interpretation
FILL FROM SCORED MANIFEST	—	—	—	Include every bound-separated pair and its case-cluster bootstrap dominance margin.

E Reproduction Setup

The released package separates checklist drafting, benchmark execution, and evidence scoring. Paths in the table are release-relative; hostnames, live API keys, and machine-specific absolute paths are not part of the public artifact.

Table 15: End-to-end reproduction pipeline. The adapter stage preserves raw benchmark evidence and native evaluator outputs; final Evidence Pass, Evidence Fail, and Unknown labels are assigned only in the scoring stage.

Stage	Reproduction details
Checklist draft	Entry point: <code>make draft</code> . Model/configuration: OpenRouter OpenAI GPT-5.4, temperature 0, high reasoning effort, 12,000-token output budget, 180-second timeout. Input: <code>case_packet.md</code> . Outputs: <code>checklist.yaml</code> , API response, LLM-call log, and reasoning summary.
Validation and locking	Entry point: offline checklist validator and case-lock update script. Inputs: <code>case_packet.md</code> and draft checklist. Outputs: validated checklist and case-lock record.
Evidence collection	Entry point: benchmark-specific adapter over the official benchmark runner. Configuration: agent settings from <code>configs/agents.yaml</code> ; no final evidence verdict. Inputs: <code>job.json</code> , benchmark installation, official assets, and agent config. Outputs: raw run record, artifact manifest, raw artifacts, native evaluator output, stdout/stderr, and call logs.
Evidence scoring	Entry point: <code>make score</code> . Model/configuration: Codex CLI GPT-5.4, xhigh reasoning effort, read-only sandbox, at most two attempts. Inputs: locked checklist and saved evidence directory. Outputs: score JSON/YAML, score manifest, stdout/stderr logs, event stream, telemetry, and reasoning summary.

Table 16: Agent-call configuration for the main experiment.

Agent	Provider	Model	Temp.	Output budget	Timeout / retry
A	OpenRouter	OpenAI GPT-5.4	0	4096	120 s / 2
B	OpenRouter	Claude Opus 4.7	0	4096	120 s / 2
C	OpenRouter	DeepSeek V4 Pro	0	4096	120 s / 2

Table 17: Machine roles used for raw evidence collection. Drafting and final scoring can run on any authenticated CPU machine and require no GPU.

Benchmarks	Machine role	Configuration	Docker
AGENTDOJO, APPWORLD, τ^3 -bench retail	General adapter VPS	8 CPU, 32 GB memory, no GPU	Yes
WEBARENA-VERIFIED	WebArena adapter VPS	4 CPU, 16 GB memory, no GPU	Yes

F Release and Result Files

Public release should make denominators and bounds auditable without exposing live credentials, real third-party keys, or side-effecting services.

- **Released publicly.** Case ids, locked case checklists, native claim, checklist rationales or source pointers, stronger-measurement flags and rationales, native label, evidence label, reason code, bound contribution, denominator audit, aggregate tables, plotting scripts, and audit instructions.
- **Released under access control when needed.** Full traces that contain prompt-injection content, synthetic personal data, or credential-shaped strings after scrubbing and sandbox re-pinning.
- **Not released.** Live credentials, real API keys, or artifacts that would reproduce side effects against non-sandboxed services.
- **Versioning.** Reported bounds are paired with a version tuple for the scorer, manifest, checklist template, checklist-generation prompt, and Unknown-reason taxonomy.

497 The final build should be produced from the scored manifest. The file
498 `outputs/latex/results_macros.tex` should set `\resultdatatrue` and define the domain-
499 level macros used in Tables 2–4. Optional row files can populate the longer appendix tables. Missing
500 values should remain visibly marked as `FILL` rather than replaced by guessed numbers.

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